CSF ICT 2 - Practice Questions and Answers

Computer Systems Fundamentals (4COSC009C) · December 2023 · typed up from test screenshots

About ICT 2

Topics covered: Process Management, Partition Sector Zero, Memory Management, Networking Theory and Calculations.

What to revise: all tutorial exercises to date and the Practice Test.

Part 1 - ICT 2 Practice Test (9 questions)

From the Blackboard practice test, December 2023. The answer key at the end of this part comes from an attempt that scored 100/100.

Question 1

Using the partition boot sector image provided with the test (Partition_Boot_Sector_1.PDF / 4COSC004W-Test2-PSZdd_Image1.pdf), answer the following:

- The partition starts at sector number **Blank 1**.
- The size of the Reserved Area is **Blank 2** sector(s).
- There are **Blank 3** FATs in this partition.
- The size of each FAT is **Blank 4** sectors.
- Therefore the start address of the Root Directory is obtained by taking the partition start address at sector **Blank 5**, then adding the size of the Reserved Area of **Blank 6** sectors and finally adding the sizes of all the FATs (which is equal to **Blank 7** sectors).
- This makes the start address of the Root Directory sector number **Blank 8**.

Question 2

In the context of this module, a quantum is a special branch of computing. True or false?

A. True
B. False

Question 3

In a round robin process management system, each process is able to access the CPU as and when needed. True or false?

A. True
B. False

Question 4

Which of the following statements are correct? (tick all the correct statements)

A. SAN stands for Storage Area Network
B. MAN stands for Metropolitan Area Network
C. UTP stands for Unshielded Twisted Pair
D. STP stands for Shielded Twisted Pair
E. LAN stands for Local Area Network

Question 5

What is the first valid host address for the network which includes the IP address 163.56.170.94/22?

Question 6

The CPU can only perform simple arithmetic. Some simple arithmetic operations are given below. (Tick the correct answers)

- **A.** Divide one number by another number
- **B.** Add one number to another number
- **C.** Multiply one number by another number
- **D.** Multiply a number by 3
- **E.** Divide a number by 2
- **F.** Multiply a number by 2
- **G.** Subtract one number from another number
- **H.** Divide a number by 3

Question 7

Given an IP address 172.16.0.1 and a subnet mask of 255.255.240.0, the network address is **Blank 1**, the broadcast address is **Blank 2** and the range of host addresses is **Blank 3** to **Blank 4**.

Question 8

Given an IP address of 10.34.23.134 and a subnet mask of 255.255.0.0, the result of performing an AND operation on these two sets of numbers is 10.34.0.0. What is this known as?

- **A.** First Valid Host Address
- **B.** Broadcast Address
- **C.** None of the listed options
- **D.** Last Valid Host Address
- **E.** Network Address

Question 9

A new process is about to be instantiated, which will require 125 kB of memory. The Operating System uses the **Best-fit** method for memory allocation. The memory map is shown below (lowest address at the bottom).

Address range	Contents	Label
above 1475 kB	Process (125 kB)	
1250 - 1475 kB	Free space (225 kB)	D
1100 - 1250 kB	Process (150 kB)	
700 - 1100 kB	Free space (400 kB)	C
575 - 700 kB	Process (125 kB)	
425 - 575 kB	Free space (150 kB)	B
300 - 425 kB	Process (125 kB)	
100 - 300 kB	Free space (200 kB)	A
0 - 100 kB	Operating System (100 kB)	

Which free space will the Operating System allocate to the new process?

- **A.** None - the OS will not find a location in memory to allocate to the new process
- **B.** D
- **C.** A
- **D.** C
- **E.** B

Answers (verified - 100/100 attempt)
Q1: 1048576, 2, 2, 7, 2048, 2, 14, 2064 (2048 + 2 + 14 = 2064)
Q2: False Q3: False Q4: all five statements (A-E) are correct
Q5: 163.56.168.1
Q6: all eight options were ticked in the 100/100 attempt
Q7: network 172.16.0.0, broadcast 172.16.15.255, hosts 172.16.0.1 to 172.16.15.254
Q8: E - Network Address
Q9: E - B (150 kB is the smallest free space that fits 125 kB)

Part 2 - Networking Quiz (11 questions)

Blackboard networking quiz, December 2023. Answers were worked out from the module material; they match the options selected in the screenshots.

Question 1

Which of the following statements are correct? (tick all the correct statements)

- **A.** UTP stands for Unshielded Twisted Pair
- **B.** STP stands for Standard Twisted Pair
- **C.** MAN stands for Metropolitan Area Network
- **D.** LAN stands for Local Area Network
- **E.** SAN stands for Storage Access Network

Question 2

This question concerns a closed network. Which of the following statements concerning Network Hubs and Switches are correct?

- **A.** Hubs send messages received to all Host machines connected to the Hub
- **B.** Switches send messages received to all Host machines connected to the Switch
- **C.** Hubs only send messages received to the specified destination Host machine.
- **D.** *(text of this option not visible in the screenshot)*

Question 3

What special piece of equipment is used in bus topology?

- **A.** a terminator
- **B.** a switch
- **C.** a hub
- **D.** a router

Question 4

In a modern office environment, the computers in a network which has a star topology are usually connected using one of these. Select the correct one.

- **A.** switch
- **B.** hub
- **C.** router
- **D.** bridge

Question 5

Why is redundancy needed in some types of networks or applications?

- **A.** to ensure against system failure
- **B.** to increase complexity so that IT professionals have more work

C. to ensure that the computer systems run efficiently
D. to double check that employees are not stealing money or data from the company

Question 6

What does NIC stand for?

A. network interconnect card
B. network interface card
C. network interfacing card
D. network interconnecting card
E. network inclusive card

Question 7

What does RAM stand for?

A. Random Access Memory
B. Real Access Memory
C. Random Archive Memory
D. Real Active Memory

Question 8

What does ROM stand for?

A. Read Only Memory
B. Real Optical Memory
C. Random Optical Memory
D. Real On-board Memory

Question 9

What is the term used to describe the topology for the way that cables are laid in a network?

A. Physical Topology
B. Logical Topology
C. Topology Hardware
D. Network Topology

Question 10

What is the term used to describe the topology for the way that computers access network media?

A. Physical Topology
B. Logical Topology
C. Topology Hardware
D. Network Topology

Question 11

In computing, a Quantum is:

A. A time interval
B. A type of CPU
C. A method of process management
D. None of the listed options

Answers

Q1: A, C, D (STP is *Shielded* Twisted Pair; SAN is Storage *Area* Network)

Q2: A (and D, whose text is not visible). Feedback given: hubs send messages to all machines in the network, which causes excessive communication and collisions; switches only send the message to the specified destination host, giving a point-to-point connection (a collision domain of only host machines, so no collisions can occur).
Q3: A Q4: A Q5: A Q6: B Q7: A Q8: A Q9: A Q10: B Q11: A

Part 3 - IP Subnetting Quizzes

Quiz A (5 questions, 1 point each)

Question 1

Given an IP address 110.32.5.115 and a subnet mask of 255.255.252.0, the network address is **Blank 1**, the broadcast address is **Blank 2** and the range of host addresses is **Blank 3** to **Blank 4**.

Question 2

The host IP address range for 192.168.36.18/30 is **Blank 1** to **Blank 2**.

Question 3

What is the first valid host on the subnetwork that the node 192.168.192.219 with a subnet mask of 255.255.255.224 belongs to?

Question 4

The last valid host on the subnetwork 172.21.108.0 with subnet mask 255.255.254.0 is **Blank 1**.

Question 5

Which subnet does host 172.22.14.13/24 belong to?

Quiz B - classful network masks

Question 1

In this question, the network mask is classful. For the network 192.168.15.0/28, the maximum number of subnets is **Blank 1** and the maximum number of hosts per subnet is **Blank 2**.

Question 2

Same as above for the network 192.168.15.0/27.

Question 3

Same as above for the network 172.168.15.0/27.

Answers (worked out)
A1: /22, block size 4 in the 3rd octet: network 110.32.4.0, broadcast 110.32.7.255, hosts 110.32.4.1 to 110.32.7.254
A2: /30, block size 4: 192.168.36.17 to 192.168.36.18
A3: /27, block size 32 (192-223): first host 192.168.192.193
A4: /23 covers 172.21.108.0 - 172.21.109.255: last host 172.21.109.254
A5: 172.22.14.0
B1: class C (default /24), 4 bits borrowed: 2^4 = 16 subnets, 2^4 - 2 = 14 hosts (confirmed in the screenshot)
B2: 3 bits borrowed: 8 subnets, 30 hosts
B3: class B (default /16), 11 bits borrowed: 2048 subnets, 30 hosts

Part 4 - FAT-16 Partition Boot Sector Exercises

Using the data shown in the partition boot sector file, complete the following statements:

1. the number of bytes per sector is ___
2. the number of sectors per cluster is ___
3. the size in sectors of the Reserved Area is ___
4. the number of FATs is ___
5. the size in sectors of the partition is ___ and
6. the size in sectors of each FAT is ___

The number of sectors before the start of the partition is ___. Does this value match the offset in the disk of the start of the partition? (yes / no)

Armed with the data from your calculations above: the first FAT starts at sector ___, the second FAT starts at sector ___ and the Root Directory starts at sector ___.

Answers (marked correct on Blackboard)

	fat-16-1.dd	**fat-16-2.dd**
Bytes per sector	512	512
Sectors per cluster	16	32
Reserved Area (sectors)	1	2
Number of FATs	2	2
Partition size (sectors)	984456	1957856
Size of each FAT	241	239
Sectors before partition	632 (yes)	32 (yes)
First FAT starts at	633	34
Second FAT starts at	874	273
Root Directory starts at	1115	512